

Lecture 7: Balance Property, Auto-Calibration & Lift Charts

Deep Learning for Actuarial Modeling
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ABSTRACT

This lecture discusses unbiasedness, the balance property and auto-calibration which all refer to model calibration. These are important properties in financial applications to ensure that prices are correct on average. Moreover, lift charts are illustrated which are a popular actuarial tool for model selection.

1 Unbiasedness and the balance property

Generally, *unbiasedness* is an important property in actuarial pricing, regardless of what specific meaning one underpins unbiasedness.

- An *in-sample bias* needs to be avoided in model selection, otherwise the predictor generalizes poorly to new data.
- An estimated regression model should be void of a *statistical bias* to ensure that the average price level is correctly specified.
- Most regression models include an intercept; see GLM lecture. This intercept is called *bias term* in the machine learning community.
- There is some concern about unfair discrimination in algorithmic decision making. Any kind of unfair treatment of individuals or groups with similar features is related to a *bias*, coined *unfair discrimination bias*.

1.1 Global statistical unbiasedness

- An estimated model $\hat{\mu}_{\mathcal{L}}$, being fitted on a learning sample $\mathcal{L} = (Y_i, \mathbf{X}_i, v_i)_{i=1}^n$, is *globally statistically unbiased* if

$$\mathbb{E} [v\hat{\mu}_{\mathcal{L}}(\mathbf{X})] = \mathbb{E}[vY],$$

assuming that $(Y, \mathbf{X}, v)$ is independent of $\mathcal{L}$.

- This unbiasedness is out-of-sample: $(Y, \mathbf{X}, v)$ vs. $\mathcal{L}$.
- Verification of unbiasedness needs knowledge of the mean $\mathbb{E}[vY]$; and the possibility to re-sample $\mathcal{L}$ and $(\mathbf{X}, v)$ for an empirical verification.
- There are various conditional versions, e.g., conditional on $\mathcal{L}$.

1.2 The balance property

- **Definition.** A regression model *fitting procedure* $\mathcal{L} \mapsto \hat{\mu}_{\mathcal{L}}$ satisfies the *balance property* if for almost every realization of the learning sample $\mathcal{L} = (Y_i, \mathbf{X}_i, v_i)_{i=1}^n$, the following identity holds

$$\sum_{i=1}^n v_i \hat{\mu}_{\mathcal{L}}(\mathbf{X}_i) = \sum_{i=1}^n v_i Y_i.$$

- The balance property is an *in-sample property* that is easy to verify.
- It was introduced in the context of credibility theory by Bühlmann and Gisler (2005); see also Lindholm and Wüthrich (2025).
- It does not require the knowledge of the true mean $\mathbb{E}[vY]$.
- The balance property is a *re-allocation* of the total portfolio claim.
- This can also be interpreted in the light of posterior risk sharing.

-
- **Mathematical result.** MLE estimated GLMs using the canonical link comply with the balance property (if and only if w.r.t. the canonical link choice).

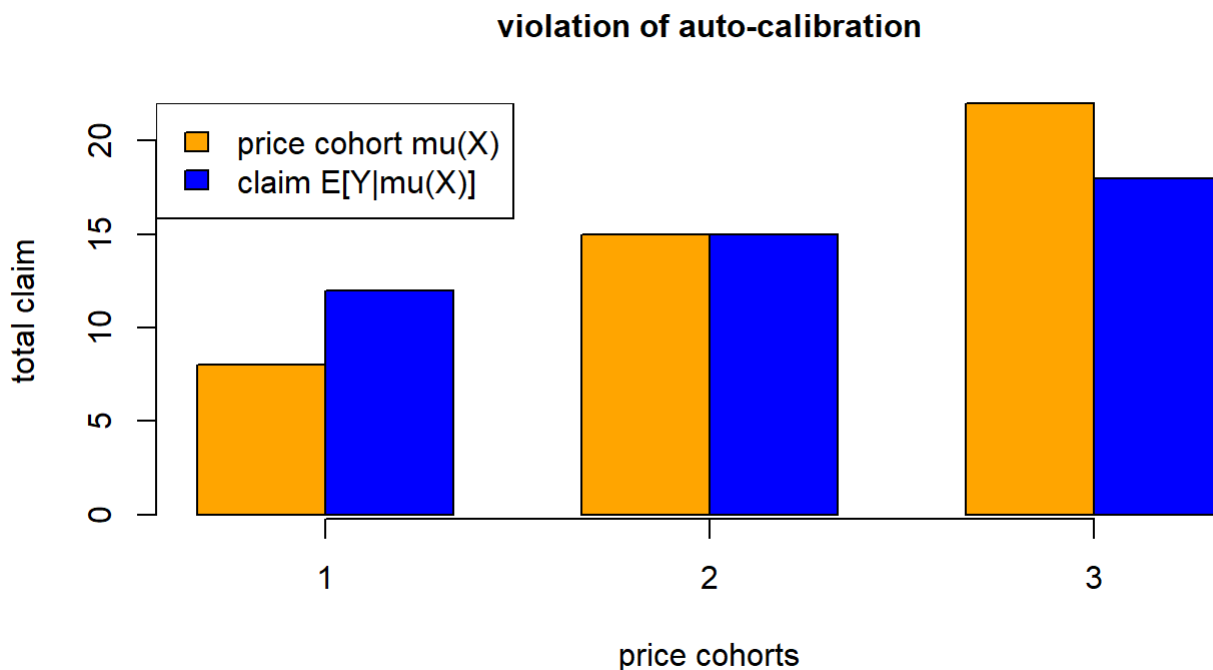
- If the balance property is not fulfilled, a correction should be applied. In most cases, one adjusts the bias term/intercept correspondingly.
- A correction may also be necessary if the future claims level changes, e.g., because of non-stationarity and/or inflation; see Brauer, Menzel and Wüthrich (2025) on concept drift.
- Under the canonical link choice, there are different ways to rectify the balance property within neural networks; see notebooks of Wüthrich *et al.* (2025).

2 Auto-calibration

- **Definition.** A regression function $\mu : \mathcal{X} \rightarrow \mathbb{R}$ is *auto-calibrated* for $(Y, \mathbf{X})$ if, a.s.,

$$\mu(\mathbf{X}) = \mathbb{E}[Y | \mu(\mathbf{X})].$$

- Auto-calibration implies that every price cohort $\mu(\mathbf{X})$ is on average self-financing for its corresponding claims Y .
- Auto-calibrated pricing schemes avoid *systematic cross-financing*.
- The following example shows a violation of auto-calibration.



- Price cohort 1 is systematically cross-financed by price cohort 3.

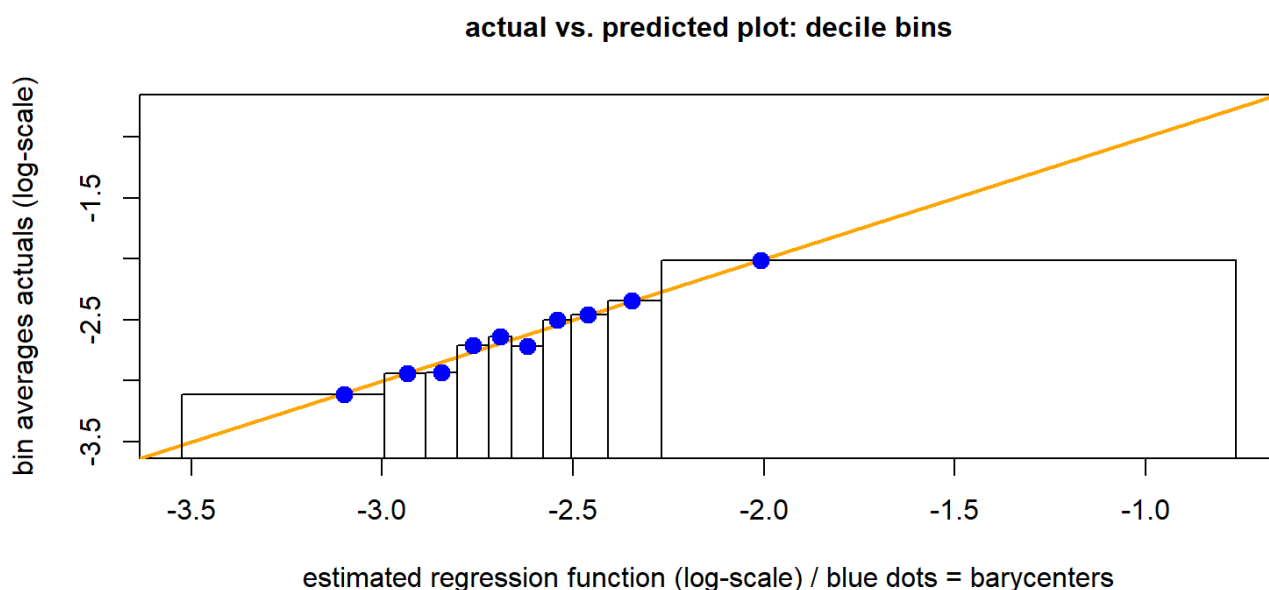
2.1 MTPL GLM example, revisited: auto-calibration

```
# we construct an actual vs. predicted plot - by decile binning

# decile binning (out-of-sample on test data)
test$freq <- test$GLM/test$Exposure # GLM predictor from previous
  lecture
qq <- quantile(test$freq, probs = c(0:10)/10)
# allocation all instances to deciles
test$qq <- 1
for (t0 in 2:10){test$qq <- test$qq + as.integer(test$freq>qq[t0])}
# compute empirical means on deciles
dd <- data.frame(test %>% group_by(qq) %>%
  summarize(yy = sum(ClaimNb),
    mm = sum(GLM),
    vv = sum(Exposure)))

#
dd$yy <- dd$yy/dd$vv # bin averages -> y-axis of next graph
  (actuals)
dd$mm <- dd$mm/dd$vv # bin barycenters -> x-axis (predictor averages)
```

- Actual vs. predicted plot using decile binning



- Is this auto-calibrated? Are the blue dots on the orange diagonal?
- Observe the non-monotonicity.

Actual vs. predicted plot

- The above plot is known as *actual vs. predicted plot*.
- One plots the observations (actuals) Y on the y -axis against the predictors $\hat{\mu}(\mathbf{X})$ on the x -axis.
- To reduce volatility, one performs quantile binning w.r.t. the predictors $\hat{\mu}(\mathbf{X})$, and one builds empirical means on these bins for predictors and responses.
- The rectangles show the decile bounds, the x -values of the blue dots corresponds to the barycenters of the predictors in these decile bins.
- Often, for a more meaningful plot, one plots the axes on the log-scale.
- If the regression function $\hat{\mu}(\cdot)$ has been estimated on a learning sample $\mathcal{L}$, this plot should be performed on the test sample $\mathcal{T}$.

2.2 Local regression

- The *local regression* of Loader (1999) gives a smooth version of the previous actual vs. predicted plot.
- A local regression smooths within fixed windows.
- Here, we use quadratic splines for the local regression

$$Y_i \sim X_i := \hat{\mu}^{\text{GLM}}(\mathbf{X}_i).$$

- The bandwidth $\delta(X_i)$ (window size) is chosen such that the smoothing window $\Delta(X_i)$ contains a nearest neighbor fraction of $\alpha = 10\%$ of the data.
- For technical details, see the notebooks of Wüthrich *et al.* (2025).

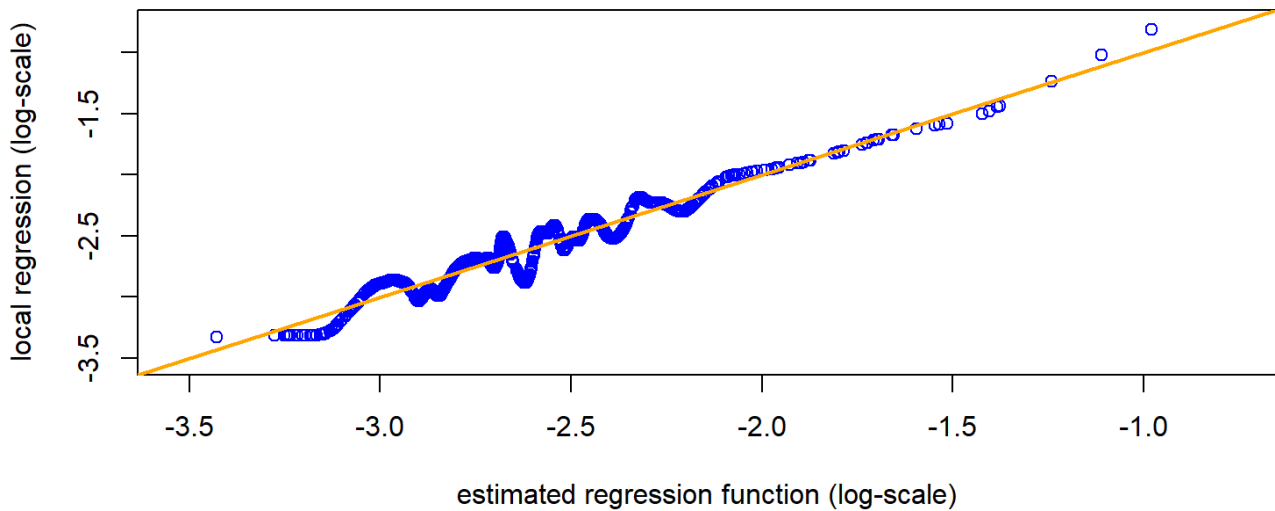
```
## we construct an actual vs. predicted plot – by a local regression
library(locfit)

### local regression approach
set.seed(100)
# select only finitely many samples for the following plot (illustration)
kk <- sample(c(1:nrow(test)), size=1000)

# local regression fit
spline0 <- predict(
  locfit(test$ClaimNb/test$Exposure ~ test$freq,
        weights=test$Exposure,
        alpha=0.1,           # nearest neighbor
        fraction,
        deg=2),              # quadratic splines
  newdata=test[kk,]$freq)
```

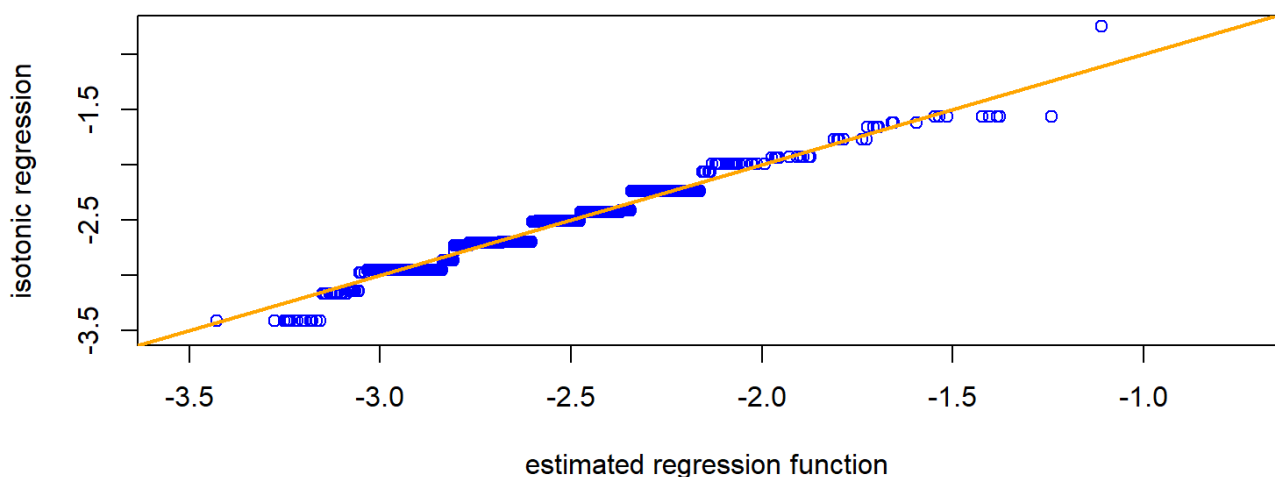
- This is illustrated in the next graph.

actual vs. predicted plot: local regression fit



- The hyper-parameters of the nearest neighbor fraction $\alpha \in (0, 1]$ and the degree of the splines have a crucial influence on the results.
- The tails seem fine, but the fluctuation in the middle are a bit large.
- *Isotonic regression* is a more robust alternative (no hyper-parameters).

actual vs. predicted plot: isotonic regression



- This is a monotone regression, not requiring hyper-parameter tuning. Over-fitting at both ends of the graph should be taken care off.

- Graph seems to support auto-calibration, except may be in the tails.
- The blue graph is (empirically) auto-calibrated! For more details, see Wüthrich and Ziegel (2024).

3 Lift charts

- *Lift charts* compare regression models.
- We select a second regression model.

```
## 2nd Poisson log-link GLM with different covariates (we drop AreaGLM,  
    and we add BonusMalusGLM and VehAgeGLM compared to the previous  
    GLM)  
  
d.glm2 <- glm(ClaimNb ~ DrivAgeGLM + BonusMalusGLM + VehBrand + VehGas +  
    DensityGLM + VehAgeGLM, data=learn, offset=log(Exposure),  
    family=poisson())  
  
## predict in-sample and out-of-sample  
learn$GLM2 <- predict(d.glm2, newdata=learn, type="response")  
test$GLM2  <- predict(d.glm2, newdata=test,  type="response")
```


3.1 In-sample and out-of-sample Poisson deviance losses

```
Poisson.Deviance <- function(pred, obs, weights){ # scaled with 100
  100*2*(sum(pred)-sum(obs)+sum(log((obs/pred)^(obs))))/sum(weights)}

# 1st GLM (in- and out-of-sample losses)
round(c(Poisson.Deviance(learn$GLM, learn$ClaimNb, learn$Exposure),
        Poisson.Deviance(test$GLM, test$ClaimNb, test$Exposure)), 3)
```

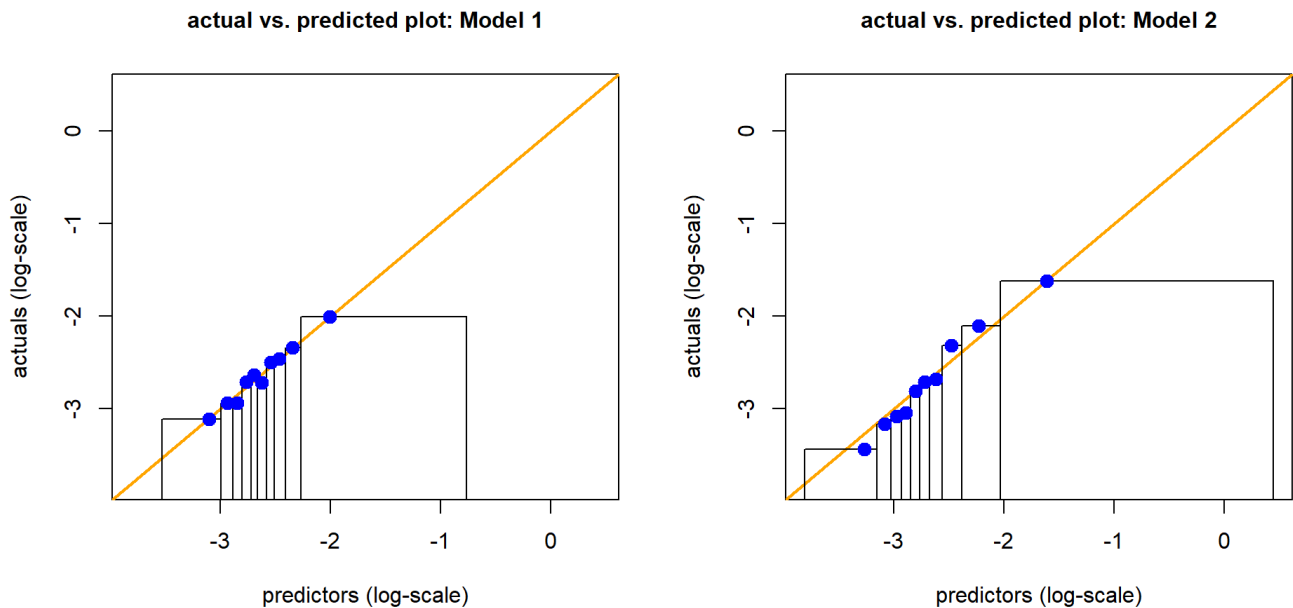
```
[1] 46.954 47.179
```

```
# 2nd GLM (in- and out-of-sample losses)
round(c(Poisson.Deviance(learn$GLM2, learn$ClaimNb, learn$Exposure),
        Poisson.Deviance(test$GLM2, test$ClaimNb, test$Exposure)), 3)
```

```
[1] 45.706 45.669
```

- It seems that the second GLM is better. Can we verify this graphically?

3.2 Actual vs. predicted plot using decile binning



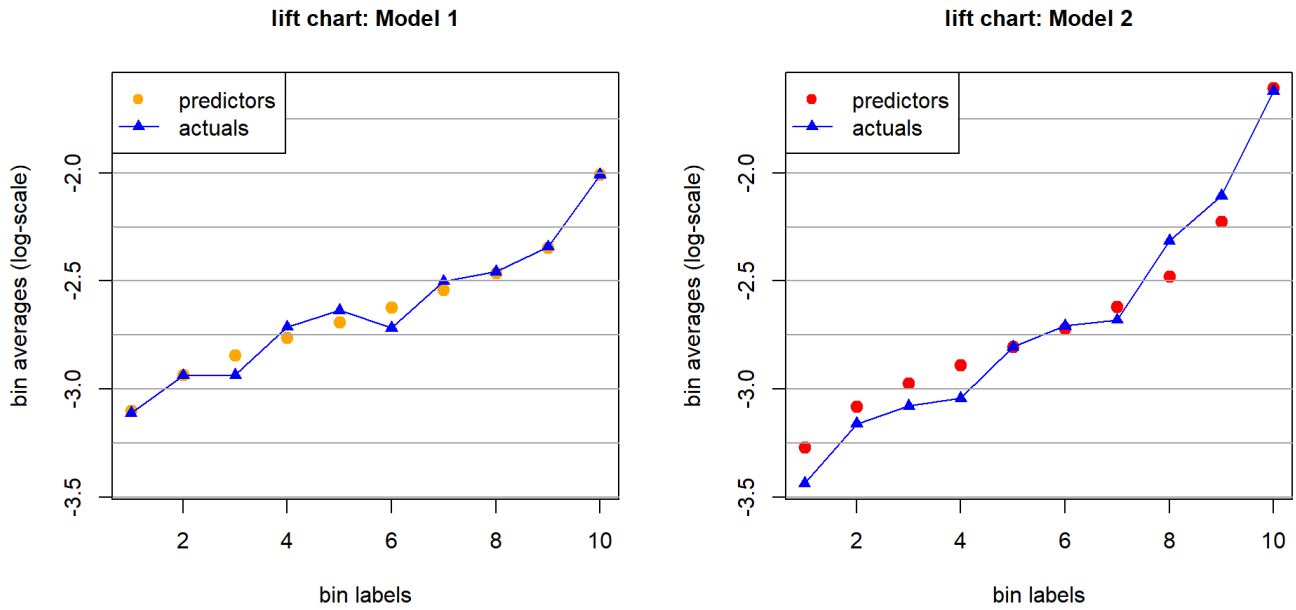
From these plots it is difficult to draw conclusions about model selection.

3.3 Lift chart

A *lift chart* shows the same statistics in a different graph: it plots the binned prediction averages and the binned response averages both on the y -axis, against the bin labels on the x -axis.

There is the following interpretation; see Goldburd *et al.* (2020).

- *Auto-calibration*. The predictors (orange circles) and the actuals (blue triangles) should approximately coincide; see next graph.
- *Monotonicity*. The actuals (blue triangles) should be monotone under auto-calibration (up to the pure noise – *correct risk ranking*).
- *Discrimination*. The better the regression model can discriminate the claims, the bigger the difference between the lowest and the highest quantiles; this is called *the lift*.



- *Auto-calibration.* Model 1 seems better auto-calibrated than Model 2.
- *Monotonicity.* Model 2 provides monotonicity, Model 1 does not.
- *Discrimination.* Model 2 has a clearly bigger lift than Model 1.

3.4 Double lift chart

- A *double lift chart* considers both regression models in the same graph.
- Assume that all predictions are strictly positive.
- Out-of-sample predictions of Model 1 and 2: $(\hat{\mu}_k(\mathbf{X}_t))_{t=1}^m$, $k = 1, 2$.
- Compute the ratios

$$\kappa_t = \frac{\hat{\mu}_2(\mathbf{X}_t)}{\hat{\mu}_1(\mathbf{X}_t)} \quad \text{for } 1 \leq t \leq m.$$

- Use the ratios $(\kappa_t)_{t=1}^m$ for quantile binning:
 - Smallest bin: Model 2 judges the risks more positively than Model 1.
 - Largest bin: Model 2 judges the risks more negatively than Model 1.

```

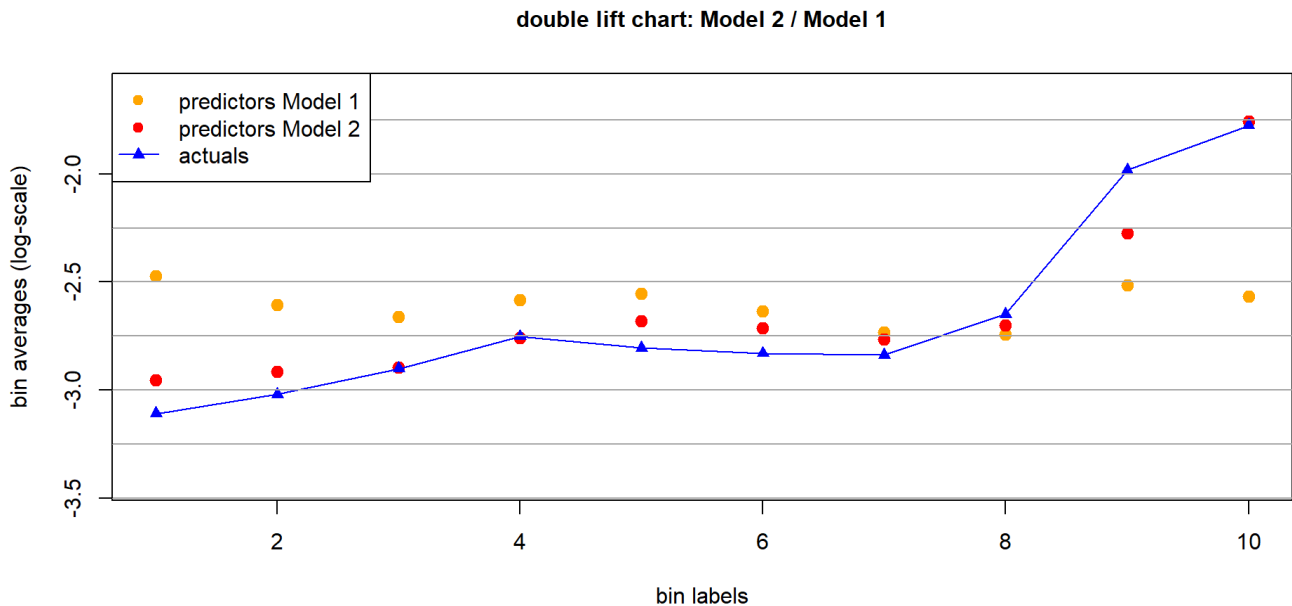
# decile binning for the double lift chart (out-of-sample on test data)
test$DL <- test$GLM2/test$GLM
qq <- quantile(test$DL, probs = c(0:10)/10)

# decile allocation of all instances
test$qq <- 1
for (t0 in 2:10){test$qq <- test$qq + as.integer(test$DL>qq[t0])}

# empirical means on all bins
dd <- data.frame(test %>% group_by(qq) %>%
  summarize(yy = sum(ClaimNb),
            mm1 = sum(GLM),
            mm2 = sum(GLM2),
            vv = sum(Exposure)))

#
dd$yy <- dd$yy/dd$vv      # bin averages actuals
dd$mm1 <- dd$mm1/dd$vv   # bin predictor GLM1
dd$mm2 <- dd$mm2/dd$vv   # bin predictor GLM2

```



- Based on this double lift chart we give preference to Model 2.
- Since both models have weaknesses, there needs to be a clearly better predictive model than both of those.

4 Appendix: Gini score

- A popular model selection tool is the *Gini (1936) score*.
- Attention, the Gini score is **not** a *strictly consistent scoring* function, thus, its minimization does not imply that we have found the best predictive model.
- The Gini score is purely *rank-based*, i.e., it can be used to assess whether the predictive model gives the *correct risk ranking*.
- For treatment of actuarial issues such as integration of case weights, we refer to Brauer and Wüthrich (2025).
- This latter reference also provides the code for the Gini Score implementation.

5 Appendix: Murphy's score decomposition

- Choose a strictly consistent loss function L for mean estimation.
- *Murphy (1973)'s score decomposition* is given by

$$\mathbb{E}[L(Y, \mu(\mathbf{X}))] = \text{UNC}_L - \text{DSC}_L + \text{MSC}_L,$$

with *uncertainty*, *discrimination (resolution)* and *miscalibration* defined by, respectively,

$$\begin{aligned}\text{UNC}_L &= \mathbb{E}[L(Y, \mu_0)] \geq 0, \\ \text{DSC}_L &= \mathbb{E}[L(Y, \mu_0)] - \mathbb{E}[L(Y, \mu_{\text{rc}}(\mathbf{X}))] \geq 0, \\ \text{MSC}_L &= \mathbb{E}[L(Y, \mu(\mathbf{X}))] - \mathbb{E}[L(Y, \mu_{\text{rc}}(\mathbf{X}))] \geq 0.\end{aligned}$$

- The positivities follow from the strict consistency.
 - We have the re-calibrated version $\mu_{\text{rc}}(\mathbf{X}) = \mathbb{E}[Y|\mu(\mathbf{X})]$.
-

```

from model_diagnostics.scoring import (
    PoissonDeviance,
    decompose,
    HomogeneousExpectileScore,
)
from model_diagnostics.calibration import compute_bias

scoring_function = PoissonDeviance()
scoring_function = HomogeneousExpectileScore(degree=1.000000001,
                                              level=0.5)
df_pred_test = pd.DataFrame({"GLM_2": test["GLM_2"], "GLM": test["GLM"]})
y_test = test["ClaimNb"] / test["Exposure"]
df = decompose(
    y_obs=y_test,
    y_pred=df_pred_test,
    scoring_function=scoring_function,
    weights=test["Exposure"],
)
df.sort(["score"])

```

- Results of Murphy's score decomposition using the Poisson deviance loss as scoring function.

model	miscalibration	discrimination	uncertainty	score
str	f64	f64	f64	f64
"GLM_2"	0.001904	0.024886	0.479674	0.456692
"GLM"	0.001078	0.008966	0.479674	0.471786

- GLM2 has a (much) better discrimination.
- GLM2 has the bigger miscalibration error that needs to be improved to avoid a systematic cross-financing within the portfolio, i.e., GLM2 has an auto-calibration issue.
- Basically, these findings confirm the conclusions from the lift charts.

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https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5162304

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References

- Brauer, A., Menzel, P. and Wüthrich, M.V. (2025) “Model monitoring: A general framework with an application to non-life insurance pricing,” *arXiv preprint* [Preprint]. Available at: <https://arxiv.org/abs/2510.04556>.
- Brauer, A. and Wüthrich, M.V. (2025) “Gini score under ties and case weights,” *arXiv preprint* [Preprint]. Available at: <https://arxiv.org/abs/2511.15446>.
- Bühlmann, H. and Gisler, A. (2005) *A course in credibility theory and its applications*. Springer. Available at: <https://doi.org/10.1007/3-540-29273-X>.
- Gini, C. (1936) “On the Measure of Concentration with Special Reference to Income and Statistics,” *Colorado College Publication*, 208, pp. 73–79.
- Goldburd, M. *et al.* (2020) *Generalized linear models for insurance rating*. 2nd ed. Casualty Actuarial Society (CAS monograph series, 5). Available at: <https://www.casact.org/sites/default/files/2021-01/05-Goldburd-Khare-Tevet.pdf>.

- Lindholm, M. and Wüthrich, M.V. (2025) "The balance property in insurance pricing," *Scandinavian Actuarial Journal* [Preprint]. Available at: <https://doi.org/10.1080/03461238.2025.2552909>.
- Loader, C. (1999) *Local regression and likelihood*. Springer. Available at: <https://doi.org/10.1007/b98858>.
- Murphy, A.H. (1973) "A new vector partition of the probability score," *Journal of Applied Meteorology and Climatology*, 12(4), pp. 595–600. Available at: https://journals.ametsoc.org/view/journals/apme/12/4/1520-0450_1973_012_0595_anvpot_2_0_co_2.xml.
- Wüthrich, M.V. *et al.* (2025) "AI Tools for Actuaries," *SSRN Manuscript* [Preprint]. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5162304.
- Wüthrich, M.V. and Ziegel, J. (2024) "Isotonic recalibration under a low signal-to-noise ratio," *Scandinavian Actuarial Journal*, 2024(3), pp. 279–299. Available at: <https://doi.org/10.1080/03461238.2023.2246743>.